

1 Symmetric Functions

1.1 The Ring of Symmetric Functions First we review the construction of the ring of symmetric functions as an inverse limit in the category of *graded rings*. This follows [Mac95]. We will assume basic notations for partitions and (skew) tableaux following [Ful96].

Definition: For $n \geq 0$, let $\Lambda_n = \mathbb{Z}[x_1, \dots, x_n]^{\mathfrak{S}_n}$ be the ring of symmetric polynomials over $\mathbb{Z}$ in n variables. For $k \geq 0$, let Λ_n^k be the k -th graded component of Λ_n , where each x_i has degree one. Then for $m \geq n$, we have a restriction morphism $\rho_{m,n}: \Lambda_m \rightarrow \Lambda_n$ sending $x_{n+1}, \dots, x_m$ to zero and $x_1, \dots, x_n$ to themselves. Furthermore, $\rho_{m,n}$ respects the grading and so we have $\rho_{m,n}^k: \Lambda_m^k \rightarrow \Lambda_n^k$ for all $k \geq 0$ and all $m \geq n \geq 0$.

The *ring of symmetric functions* Λ has as the k -th graded component the inverse limit $\Lambda^k := \varprojlim_n \Lambda_n^k$ and $\Lambda := \bigoplus_{k \geq 0} \Lambda^k$.

Note that this limit is over *graded rings* and so we do not have elements such as $\prod_{i=1}^{\infty} (1 + x_i)$. To obtain the ring of symmetric functions Λ_A over rings A other than $\mathbb{Z}$, repeating the above construction gives $\Lambda_A \simeq \Lambda \otimes_{\mathbb{Z}} A$.

1.2 Common Bases for the ring of Symmetric Functions We describe some basic examples of symmetric functions.

Definition:

- Let $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0)$ be any partition with length $\ell(\lambda) \leq n$. Then the *monomial symmetric function* $m_\lambda(x_1, \dots, x_n)$ is defined by

$$m_\lambda(x_1, \dots, x_n) = \sum_{\alpha} x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

where the summation is taken over all permutations of λ . We have that Λ^k is the free $\mathbb{Z}$ -module of rank $p(k)$ with basis $m_\lambda(x_1, \dots, x_n)$ for all $n \geq k$ as $\lambda \vdash k$ ranges over all partitions of k , where $p(k)$ is the number of partitions of k .

- For $r \geq 0$, the r -th *elementary symmetric function* e_r is defined by

$$e_r = \sum_{i_1 < \dots < i_r} x_{i_1} \cdots x_{i_r} = m_{(1^r)}.$$

For a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ we define $e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$. The *Fundamental Theorem of Symmetric Functions* states that over finitely many variables, say $x_1, \dots, x_n$, the elementary symmetric functions $e_1, \dots, e_n$ are algebraically independent and $\Lambda_n \simeq \mathbb{Z}[e_1, \dots, e_n]$. For a clean proof of this fact due to van der Waerden, see [Stu08, Theorem 1.1.1].

- For $r \geq 0$, the r -th *complete homogeneous symmetric function* h_r is defined by

$$h_r = \sum_{i_1 \leq \dots \leq i_r} x_{i_1} \cdots x_{i_r} = \sum_{\lambda \vdash r} m_\lambda.$$

For a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ we define $h_\lambda = h_{\lambda_1} h_{\lambda_2} \cdots$.

- For $r \geq 0$, the r -th *power sum symmetric function* p_r is defined by

$$p_r = \sum_{i \geq 1} x_i^r = m_{(r)}.$$

For a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ we define $h_\lambda = h_{\lambda_1} h_{\lambda_2} \cdots$.

1.3 Schur Functions We follow [Sta23, Chapter 7] for the exposition on Schur functions.

Definition: For a Young tableaux T , its *type* is the vector $\alpha = (\alpha_1, \alpha_2, \dots)$ where α_i is the number of entries of T labeled with i . If T has type α , write $\bar{x}^T = x_1^{\alpha_1} x_2^{\alpha_2} \dots$. For a partition λ , the *Schur function* $s_\lambda(\bar{x})$ is defined by

$$s_\lambda(\bar{x}) = \sum_{\text{SSYT } T \text{ of shape } \lambda} \bar{x}^T$$

where the summation is taken over all semistandard Young tableau of shape λ .

Note that the Schur functions also form a basis of Λ .

Proposition: Let

$$a_\delta = \det(x_i^{n-j})_{1 \leq i, j \leq n} = \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

be the *Vandermonde determinant*. For a partition $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0)$, the *alternant* $a_{\lambda+\delta}$ is the determinant

$$a_{\lambda+\delta} = \det(x_i^{\lambda_j+n-j})_{1 \leq i, j \leq n}.$$

Then $s_\lambda(x_1, \dots, x_n) = \frac{a_{\lambda+\delta}}{a_\delta}$.

Proposition: For a partition λ , we have that

$$s_\lambda(x_1, \dots, x_n) = \det(h_{\lambda_i+j-i}(\bar{x}))_{1 \leq i, j \leq n},$$

$$s_{\lambda'}(x_1, \dots, x_n) = \det(e_{\lambda_i+j-i}(\bar{x}))_{1 \leq i, j \leq n},$$

the *Jacobi-Trudi* and *Nägelsbach-Kostka* identities respectively.

In particular we have that $e_r = s_{(1^r)}$ and $h_r = s_{(r)}$.

Theorem (Pieri Rules): For a partition λ and an integer $r \geq 0$, we have that

$$h_r(\bar{x}) \cdot s_\lambda(\bar{x}) = \sum_{\mu} s_\mu$$

where the summation over μ is over all partitions μ formed by added r boxes to the diagram of λ such that no two boxes are added to the same column.

We also have that

$$e_r(\bar{x}) \cdot s_\lambda(\bar{x}) = \sum_{\mu} s_\mu$$

where the summation over μ is over all partitions μ formed by added r boxes to the diagram of λ such that no two boxes are added to the same row.

1.4 An Involution and Inner Product

Definition: We have a map $\omega: \Lambda \rightarrow \Lambda$ given by $\omega(e_\lambda) = h_\lambda$.

Proposition: One can check that $\omega(h_\lambda) = e_\lambda$ so that ω is an involution. Furthermore, we have that $\omega(p_\lambda) = (-1)^{|\lambda|-\ell(\lambda)} p_\lambda$ and $\omega(s_\lambda) = s_{\lambda'}$.

Definition: We define an inner product $\langle \cdot, \cdot \rangle$ on Λ by making the two bases $\{m_\lambda\}_\lambda$ and $\{h_\mu\}_\mu$ dual: $\langle m_\lambda, h_\mu \rangle = \delta_{\lambda, \mu}$ for all partitions λ, μ .

Proposition: Let $z_\lambda = \prod_k k^{n_k(\lambda)} \cdot n_k(\lambda)!$ where $n_k(\lambda)$ is the number of parts of λ of size k . Then $\langle p_\lambda, p_\mu \rangle = z_\lambda \delta_{\lambda, \mu}$. We also have that $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda, \mu}$ so that the Schur functions form an orthonormal basis of Λ .

Proposition: The involution ω is self-adjoint: $\langle f, \omega(g) \rangle = \langle \omega(f), g \rangle$. Thus, we have that $\langle \omega(f), \omega(g) \rangle = \langle f, \omega^2(g) \rangle = \langle f, g \rangle$ for all $f, g \in \Lambda$.

1.5 Skew-Schur Functions and Skewing

Definition: For two partitions λ, μ , the *skew Schur function* $s_{\lambda/\mu}(\bar{x})$ is defined by

$$s_{\lambda/\mu}(\bar{x}) = \sum_{\text{SSYT } T \text{ of shape } \lambda/\mu} \bar{x}^T$$

where the summation is taken over all semistandard Young tableau of shape λ/μ .

Proposition: For partitions λ, μ , we have that

$$\begin{aligned} s_{\lambda/\mu}(x_1, \dots, x_n) &= \det \left(h_{\lambda_i - \mu_j + j - i}(\bar{x}) \right)_{1 \leq i, j \leq n}, \\ s_{\lambda'/\mu'}(x_1, \dots, x_n) &= \det \left(e_{\lambda_i - \mu_j + j - i}(\bar{x}) \right)_{1 \leq i, j \leq n}. \end{aligned}$$

Definition: For any symmetric function $f \in \Lambda$, the *skewing operator* $f^\perp : \Lambda \rightarrow \Lambda$ is defined by the property that for all $g, h \in \Lambda$ we have that $\langle f \cdot g, h \rangle = \langle g, f^\perp(h) \rangle$.

The discussion on skewing follows [GR26, Chapter 2].

Example: We have that $s_\mu^\perp(s_\lambda) = s_{\lambda/\mu}$.

Example: We have “dual versions” of the Pieri rules. For a partition λ and an integer $r \geq 0$, we have that

$$h_r^\perp(s_\lambda(\bar{x})) = \sum_{\mu} s_\mu$$

where the summation over μ is over all partitions μ such that λ/μ is a *horizontal strip* meaning no two boxes in the skew diagram are in the same column. We also have that

$$e_r^\perp(s_\lambda(\bar{x})) = \sum_{\mu} s_\mu$$

where the summation over μ is over all partitions μ such that λ/μ is a *vertical strip* meaning no two boxes in the skew diagram are in the same row.

References

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